General Relations for Electric and Magnetic Fields in Waveguides

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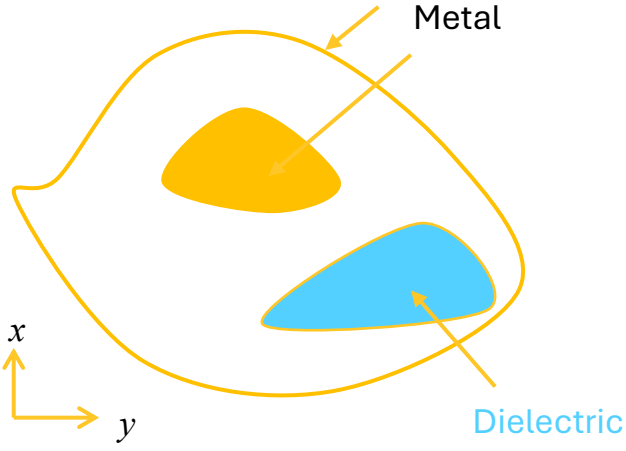
Lecture Objective

By the end of this lecture, you will be prepared to:

- Analyze waveguiding structures using Helmholtz equations and the relationships derived from Maxwell's curl equations.

Uniform Waveguiding Structures

- Consider a general uniform waveguiding structure.
- The cross-section of the structure does not change over the length of the structure (as a function of z).
- The structure comprises <u>at least</u> one conductor <u>or</u> one dielectric material, and possibly additional conductors and dielectrics.
- Theoretically, structures in the cross-section can be infinite in extent (e.g. parallel plate waveguide).



Cross-section of arbitrary waveguide

2. Uniform Waveguiding Structures

- Assume the following phasor electric and magnetic fields inside the waveguide:

$$\tilde{\vec{E}}(x,y,z) = \tilde{\vec{e}}(x,y)e^{-\gamma z} = \left(\hat{x}\,\tilde{e}_x(x,y) + \hat{y}\,\tilde{e}_y(x,y) + \hat{z}\,\tilde{e}_z(x,y)\right)e^{-\gamma z}$$

$$\tilde{\vec{H}}(x,y,z) = \tilde{\vec{h}}(x,y)e^{-\gamma z} = \left(\hat{x}\,\tilde{h}_x(x,y) + \hat{y}\,\tilde{h}_y(x,y) + \hat{z}\,\tilde{h}_z(x,y)\right)e^{-\gamma z}$$

- That is, the fields are those of an electromagnetic wave propagating in the +z-direction with propagation constant given by

$$\gamma = \alpha + j\beta$$

Some Terminology

- "Transverse" components: E_x, E_y, H_x and H_y.
- "Longitudinal" components: E_z and H_z.
- "Transverse" variation:

$$\tilde{\vec{e}}(x, y)$$

$$\tilde{\vec{h}}(x, y)$$

- "Longitudinal" variation:

$$e^{-\gamma z}$$

3. Uniform Waveguiding Structures

- For a given uniform waveguide, we need to obtain the electromagnetic ‘modes’ that can be supported by the geometry.
- These modes are determined by setting up and solving an ***eigenvalue*** problem (EVP) for the fields that can “fit” inside the cross-section of the waveguide, that is, for the fields that satisfy both Maxwell’s equations and boundary conditions.
- Let’s see how to obtain the requisite EVP …

(3D) Vector Helmholtz Equations

- Inside the waveguide, the phasor electric and magnetic fields must satisfy the vector Helmholtz equations and boundary conditions:

$$\nabla^2 \tilde{\vec{E}} + k^2 \tilde{\vec{E}} = 0$$

$$\nabla^2 \tilde{\vec{H}} + k^2 \tilde{\vec{H}} = 0$$

$$+ \text{BCs}$$

Eliminating the (Known) Longitudinal Variation

- Notice that:

$$\nabla^2 \tilde{\vec{E}} = \left(\nabla_t^{\,2} + \frac{\partial^2}{\partial z^2} \right) \tilde{\vec{e}}\, e^{-\gamma z} = \left(\nabla_t^{\,2} + \gamma^2 \right) \tilde{\vec{e}}\, e^{-\gamma z}$$

$$\nabla^2 \tilde{\vec{H}} = \left(\nabla_t^{\,2} + \frac{\partial^2}{\partial z^2} \right) \tilde{\vec{h}}\, e^{-\gamma z} = \left(\nabla_t^{\,2} + \gamma^2 \right) \tilde{\vec{h}}\, e^{-\gamma z}$$

(2D) Vector Helmholtz Equations

- Substituting into the Helmholtz equations yields:

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{\vec{e}} = 0$$

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{\vec{h}} = 0$$

$$+\ \mathrm{BCs}$$

$$k_c^{\,2} = k^2 + \gamma^2$$

(2D) Vector Helmholtz Equations

- The above two vector Helmholtz equations yield a set of six scalar Helmholtz equations (one for each scalar component of electric and magnetic field).

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_x = 0$$

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_y = 0$$

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_z = 0$$

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{h}_x = 0$$

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{h}_y = 0$$

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{h}_z = 0$$

(2D) Helmholtz Equations for Longitudinal Components

- We only need two of the six equations: the other four field components can be obtained from the two selected using Maxwell's curl equations.
- By convention, the two components usually determined via the Helmholtz equations are E_z and H_z.

$$\boxed{\begin{aligned} &\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_z = 0 \\ &\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{h}_z = 0 \end{aligned}} \qquad (1)$$

Relationships Between Transverse Field Components and Longitudinal Field Components

- These relationships follow from Maxwell's curl equations.

$$\boxed{\begin{aligned}
\tilde{e}_x &= -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{e}_z}{\partial x} + j\omega\mu\frac{\partial\tilde{h}_z}{\partial y}\right)\\
\tilde{e}_y &= -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{e}_z}{\partial y} - j\omega\mu\frac{\partial\tilde{h}_z}{\partial x}\right)\\
\tilde{h}_x &= -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{h}_z}{\partial x} - j\omega\varepsilon\frac{\partial\tilde{e}_z}{\partial y}\right)\\
\tilde{h}_y &= -\frac{1}{k_c^{\,2}}\left(\gamma\frac{\partial\tilde{h}_z}{\partial y} + j\omega\varepsilon\frac{\partial\tilde{e}_z}{\partial x}\right)
\end{aligned}} \qquad (2)$$

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